

Thirty-seven clauses. Thirty-one met in full.

Problem statement 26013 is read here as what it is — a specification. Every noun it names is a clause, every clause has a status, and the six that are not met in full say so in their own row rather than in a footnote. The partial ones are the interesting half.

Statement Automated Integration and Intelligent Harmonisation of Multi-source Geospatial Data for Urban Land Record Management

Organisation Ministry of Rural Development · DoLR

Evidence out/chennai_metro/metrics.json, run d41789a8

37

clauses extracted from
the statement

31

met in full, with
evidence

6

partial, and stated as
partial

0

claimed without
something to point at

METHOD

Re-verified from source, today

Nothing in this record is quoted from an earlier document. Each clause was checked against one of three things: a module in the shipped tree, a figure in the run record the pipeline wrote, or an endpoint that answers. Where the check was a judgement rather than a fact — "is this *enough* to count as met?" — the row is marked partial and the reasoning is in it.

The tree as counted today: **69 Python modules, 18,801 lines** across `backend/GeovaX/`, **130 test functions** in seven test modules, and **40 HTTP endpoints** including the five OGC API - Features routes.

CLAUSE GROUP 1 OF 5 · FOUR VERBS

The platform definition

"An AI-enabled geospatial integration platform capable of **automatically integrating, harmonizing, validating, and synchronizing** multiple land-related datasets with AI-generated feature extraction outputs." Four verbs, and they are not synonyms.

CLAUSE	STATUS	WHAT ANSWERS IT
Integrating	met	Five configured layers ingested and put on one datum in a single run — 14,253 + 1,826 parcels, 55,810 + 70,776 footprints. <code>ingest</code> , <code>crs.engine</code> .
Harmonizing	met	106,077 entities assembled from 558,294 claims across sources; 36,987 conflicts adjudicated, 42.7% closed automatically. <code>matching</code> , <code>conflict</code> .
Validating	met	15 topology rules with bounded, refusable repair; six-dimension confidence graded A–E on every feature; hash-chained ledger, 20 entries, chain verified. <code>topology</code> , <code>confidence</code> , <code>core.ledger</code> .
Synchronizing	partial	<code>POST /api/subscriptions</code> and the delivery log are implemented and the AOI / feature-class / change-type / confidence filters exist. Webhook delivery has not been exercised against a second real department — until it is, this is a loop-back test, and we say so.

Why "validating" is the one to press us on

Most submissions read validate as "check the geometry is valid". Here it means the platform can be *audited*: every resolution names the claims it superseded, carries a plain-language rationale, and is anchored in a Merkle root a citizen can check without trusting the department. Run `geovax verify --out out/chennai_metro` and it either reports chain intact over 20 entries or names the exact index where it breaks.

CLAUSE GROUP 2 OF 5 · TEN DATASETS

Every input the statement names

An input with no real dataset behind it is an input with no evidence. Each row names the actual data used, not a connector that would accept it.

NAMED INPUT	STATUS	REAL DATA BEHIND IT
Drone imagery	met	Real UAV flight from the OpenDroneMap UAVArena corpus, three independent photogrammetric reconstructions (ODM 3.0.0, Pix4D 4.4.10, Metashape 1.5.2). <code>ingest.tiles</code> .

NAMED INPUT	STATUS	REAL DATA BEHIND IT
Orthorectified imagery (ORI)	met	ORI rebuilt from published XYZ pyramids into georeferenced COGs at 0.051 m ground resolution ; co-registration measured between reconstructions. <code>raster.cog</code> , <code>raster.coreg</code> .
DSM / DTM datasets	met	Full DSM → DTM → nDSM → structures chain on a real calibrated float surface (Geobasis NRW DOM1, 1 m grid, orthometric). nDSM median 0.023 m — the derived ground sits on the ground. <i>Stated caveat: no Indian float DSM is openly available at this precision, and the platform's trust gate refused to treat a recovered colour ramp as elevation rather than invent one.</i>
Existing cadastral maps	met	Two independent real compilations: TNGIS statewide cadastre (6.02 M parcels) and NCSCM coastal cadastre (220,668). Both CC0-1.0. <code>ingest.vector</code> .
Revenue records	met	Survey numbers, subdivision, LGD hierarchy and FMB flags carried as canonical attributes and used as join keys — <code>kide</code> → <code>survey_number</code> resolved by value profiling alone. <i>Caveat: attribute-level, from the cadastral layers; no live RoR/patta database was available to connect to.</i>
Municipal GIS layers	met	964,053 Greater Chennai Corporation footprints with ward, zone, locality, street; 55,810 in the AOI. Automatic crosswalk mapped 6 of 13 columns. <code>attributes.schema_match</code> .
Utility network data	met	Chennai Metrowater transmission network as a real source, with dangle, undershoot and connectivity rules (<code>dangle_max_length_m 3.0</code> , <code>undershoot_tol_m 1.0</code>). <code>topology.validate</code> .
Ground truthing (GT)	partial	The GCP model ladder and Baarda iterative data snooping are implemented and tested, and CE90 / LE90 / NSSDA reporting exists. No independent ground-truth campaign was available , so the estimators are exercised on control points rather than on a real survey. This is the single largest gap in the evaluation and it is first on the deployment path.

NAMED INPUT	STATUS	REAL DATA BEHIND IT
GNSS / CORS survey data	partial	RINEX, NMEA and CSV readers are built and uncertainty is carried through to the evidence model; rule R-CTL-01 gives control precedence over every photogrammetric claim. No real CORS campaign output was obtainable — Survey of India CORS requires credentials this environment cannot get, catalogued and labelled rather than faked.
Building footprint datasets	met	Three independent real sources: GCC municipal survey, Google Open Buildings v3 (the AI-generated extraction the statement names), and AMRUT/Bhuvan. The fifth configured layer returned zero features in this AOI and the run reports that as a completeness gap.

CLAUSE GROUP 3 OF 5 · SEVEN CAPABILITIES

Everything the solution must incorporate

NAMED CAPABILITY	STATUS	MEASURED IN THIS RUN
AI/ML spatial matching	met	Weak-supervised, self-trained gradient-boosted matcher. Blocking removed 99.991% of 3.95×10^9 possible pairs; 34,697 matches accepted at mean $p = 0.999$; 5-fold CV AUC 1.0000 , calibration error 0.0002 .
Automated topology correction	met	731 repairs applied across the cadastral layers, 0 refused within tolerance ; anything beyond 5 m ² overlap or 0.5 m snap escalates instead. Mean topological confidence 0.9997 .
Intelligent attribute mapping	met	Four unseen schemas crosswalked with no configuration, on four signals. Middle-band candidates surfaced for review, not applied — <code>lgd_district_code</code> → <code>state_lgd</code> at 0.544 was correctly withheld.
Geo-referencing & coordinate transformation	met	Indian datum family (Everest 1830 / Kalianpur) with EPSG:7755 as a first-class target and EPSG:32644 as the measurement CRS; GCP ladder with blunder detection; every geometry reprojected per run. <code>crs.engine</code> , <code>crs.gcp</code> .

NAMED CAPABILITY	STATUS	MEASURED IN THIS RUN
Change detection mechanisms	met	106,648 typed discrepancies across both layer pairs, in cross-source mode rather than temporal — plus a raster change path with a measured false-positive floor from a null-change experiment.
Spatial conflict resolution framework	met	Dempster–Shafer fusion with independence discounting over 558,294 claims; conflict mass retained rather than normalised away; statutory rule engine — R-GEO-01 fired 839 times to refuse fusion beyond 10 m.
Confidence scoring for integrated outputs	met	Six explainable dimensions per feature, graded A–E. Composite mean 0.587 ; weakest dimension named in the run record (source agreement, 0.335); grade split C 51,878 / D 54,199.

The two clauses that decide whether this is a land-records platform

Conflict resolution and confidence are last in the statement and first in the design. When four sources disagree about one boundary, something must decide and something must say how sure it is. **The platform never averages.** It selects a boundary some surveyor actually observed, by evidence — because an averaged boundary is attributable to no survey, can be asked of no surveyor, and is indefensible in a dispute.

CLAUSE GROUP 4 OF 5 · SIX OUTCOMES

What the final solution is judged on

STATED OUTCOME	STATUS	EVIDENCE
Reduce manual GIS integration effort	met	106,077 features reconciled in 241.6 s on 2 vCPU , unattended, with per-stage checkpoints. The manual equivalent is a GIS analyst reconciling parcel by parcel.
Improve accuracy and consistency	met	Two undeclared systematic offsets recovered and removed (1.51 m @ 073°, 3.41 m @ 309°); topological consistency raised to 0.9997; 860 apparent mutations correctly suppressed as re-registration rather than raised.

STATED OUTCOME	STATUS	EVIDENCE
Enable seamless inter-departmental exchange	partial	OGC API - Features Part 1 is conformant and live — <code>/conformance</code> , <code>/collections</code> , <code>/collections/{id}/items</code> — so a department opens it in QGIS with no GEOVAX-specific code. The push side (subscription bus) is built but untested against a real second department , which is what "seamless" would actually require.
Accelerate cadastral finalization	met	Finalisation is blocked by contested parcels. 42.7% of 36,987 conflicts close without a human; the remaining 21,192 are ranked by decision value and batched by cause into 2 groups , so one decision generalises across hundreds of cases.
Improve interoperability	met	Open standards on every outbound path: OGC API - Features (ISO 19168-1), GeoJSON RFC 7946, CityJSON 1.1 for the LOD1 city model, Cloud Optimized GeoTIFF for raster, LGD codes as the join key.
Support standardized digital land governance	met	DILRMP-conformant 14-character ULPIN with checksum and split/merge genealogy; DPDP-Act-aware PII handling (owner attributes absent from the feature API by construction, reachable only through a purpose-bound, ledger-logged endpoint); NAKSHA specification used as the confidence target.

CLAUSE GROUP 5 OF 5 • TEN SUGGESTED TECHNOLOGIES

The suggested stack, and where we declined it

This list is latitude, not obligation — the statement calls it *suggested*. Breadth is shown where it is real, and the two declines are deliberate engineering choices with reasons.

SUGGESTED	STATUS	HOW, OR WHY NOT
Artificial Intelligence	met	Evidence-theoretic decision layer: Dempster–Shafer belief and plausibility, statutory rule engine, adjudication ranking by decision value.
Machine Learning	met	Weak supervision plus self-training over a gradient-boosted matcher; Beta posterior reliability updates from every officer decision.

SUGGESTED	STATUS	HOW, OR WHY NOT
GeoAI	met	Google Open Buildings extraction consumed as a first-class source; footprint regularisation (52.3% fewer vertices, 106 correctly refused as non-rectilinear); LOD1 extrusion.
GIS & Web-GIS	met	Operator console at <code>/map</code> plus a Next.js web-GIS client with an end-to-end Playwright suite.
Spatial databases	partial	Full PostGIS canonical schema, indexes and views (<code>db/schema.sql</code> , 443 lines) with loaders. The evaluation deliberately runs file-backed so a reviewer can reproduce it without standing up a database — so PostGIS is built and shipped, but it is not what produced these numbers.
ETL automation	met	12-stage DAG with checkpointing and resume; a crashed run resumes rather than restarting, and partial state stays inspectable.
Computer vision	met	Phase-correlation co-registration, progressive morphological ground filtering, structure extraction from height alone (328 structures, no imagery and no labels).
Cloud computing	partial	Containerised and horizontally sized from single-ULB pilot to statewide, but deliberately deployable on-premises with no external service dependency at inference time . Land records are DPDP-governed data and a district office has neither guaranteed bandwidth nor an appetite for offshore processing. Cloud-native was declined, not missed.
Spatial analytics	met	Moran's I spatial autocorrelation on residuals, CE90 / LE90 / NSSDA positional accuracy, per-ward and per-village quality rollups, development-control checks against TN CDBR 2019.
API integration frameworks	met	FastAPI serving 40 endpoints: five OGC API - Features routes plus the platform API — adjudication, provenance, lineage, subscriptions, citizen verification, GatiShakti export.

Eight things PS 26013 never asked for

Overdelivery is only worth listing if it is running. Each of these is in the shipped tree with an endpoint or a test behind it.

ULPIN genealogy	Split and merge lineage so a mutation never orphans an encumbrance history; checksum catches 100% of single-character errors. <code>core.ids</code>
Merkle-anchored ledger	Inclusion proofs let a citizen verify their own record against a gazette-published root, independently of the department. <code>GET /api/citizen/verify/{ulpin}</code>
Draft FMB generation	Geodesic polygon → G-line, perpendicular ladder offsets, F-lines → NIC CollabLand XML plus an SVG sketch in customary units, so state cadastral software can ingest it without redrawing. <code>cadastre.fmb</code> , <code>GET /api/fmb/{ulpin}</code>
Litigation hotspot mapping	Dempster–Shafer conflict mass K fused with e-Courts/NJDG dispute records and encumbrance flags into a risk index, so a Collector can aim survey budget at contested ground. <code>analytics.litigation</code>
LOD1 3-D city model	CityJSON 1.1 solids with measured height and storeys; FSI and ground-coverage checks against TN CDBR 2019 limits. <code>geoai.lod1</code>
Encroachment intelligence	Change restricted to poramboke/government land, reported as findings for verification and never as determinations. Rule R-POR-01.
Area reconciliation in customary units	Acre-cent, ground, kuzhi, guntha, kanal, with recorded-versus-geodesic discrepancy as a first-class field.
Active-learning loop	Every adjudication becomes a training example and a reliability update, so the platform's accuracy is a function of how much it has been used.

SCOPE

What this record supports, stated as it should be said

This record supports one sentence: **on 22.3 km² of central Chennai, using five real government layers and no synthetic data, the platform automatically integrated, harmonised and validated 106,077 land features in 241.6 seconds on 2 vCPU, closed 42.7% of the conflicts without a human, and published a per-feature confidence grade and an auditable reason for every decision it made.**

It does not support a claim about accuracy against ground truth, because there is no ground truth here to measure against. It does not support a claim about a second department receiving a webhook, because none has. And it does not support any claim about title.

The six partial rows are the honest half of this document

Ground truthing and GNSS/CORS are partial because the real data requires credentials this environment cannot obtain. Synchronisation and inter-departmental exchange are partial because the far end of a webhook has to be a real department. Spatial databases and cloud are partial by choice, and the choices are defended above. A conformance record with no partial rows is a marketing document.

check the three claims that need no trust

```
geovax verify --out out/chennai_metro      # chain intact over 20 entries + Merkle
curl -s localhost:8000/conformance | jq    # the OGC classes actually declared
python -m pytest tests/ -q                 # 130 test functions, seven modules
```

GEOVAX Conformance Record — clauses extracted from the published text of problem statement 26013; evidence read from `out/chennai_metro/metrics.json` (run d41789a8, 2026-09-06 16:49 UTC) and from the shipped tree. Team Perpetually Curious · SIH 2026 · Ministry of Rural Development, Department of Land Resources.